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kushagrakatiyar06.github.io/Portfolio-v2/ (Portfolio)

Top Skills

Adobe Photoshop
Video Editing
AngularJS

Languages

English (Native or Bilingual)
Hindi (Native or Bilingual)
Spanish (Elementary)

Certifications

Adobe Certified Associate in Visual Design Using Adobe Photoshop
Adobe InDesign CC
Fundamentals of Deep Learning

Honors-Awards

AP Scholar Award
AP Scholar with Distinction Award
AP Scholar with Honor Award
AP Scholar Award
Gold Seal of Biliteracy

Kushagra Katiyar

AI Intern @ Primo Brands | Computer Science & Statistics @ University of Florida | 4x Hackathon Winner
Greater Orlando

Summary

Check out my Portfolio!: https://kushagrakatiyar06.github.io/Portfolio/ Github: https://github.com/KushagraKatiyar06Devpost: https://devpost.com/kooshieHey! I am Kushagra.I am a Computer Science and Statistics student with an AI Certificate, specializing in scalable systems, full-stack development, agentic AI, and UI/UX design. My practical experience stems from corporate internships, research labs, and leading technical initiatives on campus. I have also won in 5 tracks across 4 hackathons. I want to build a career in software engineering, making speedy and scalable software. I also have an interest in incorporating generative AI in customer facing applications.Outside of the tech space, my interests include Taekwondo, Flute, Cars and Gaming.

Experience

Primo Brands
5 months

AI & Digital Experience Intern
August 2026 - Present (2 months)
Tampa, FL

AI & Digital Experience Intern
May 2026 - July 2026 (3 months)
Tampa, FL

As the first AI hire at Primo Brands, I architected and deployed a Retrieval-Augmented Generation chatbot on the company's ServiceNow knowledge base, reaching 12,000+ employees, while advising executive leadership on enterprise-wide AI adoption strategy, consisting of employee trainings and governance compliance.

• Architected a hybrid RAG chatbot combining BM25 keyword search and vector embeddings across 70+ cited ServiceNow KB articles, built on FastAPI, React, and Azure AI Search.

• Cut inference cost by 10% and response latency from 7s to 2s by stripping boilerplate via Azure Document Intelligence layouts and layering a 500-entry LRU cache at a 0.87 similarity threshold, cutting speeds by 70% compared to full retrieval passes.

• Advised the VP of Infrastructure and CIO on enterprise AI adoption strategy, informing organization-wide rollout decisions across HR, Finance, etc.

• Led office hours and hands-on trainings for 100+ concurrent attendees across HR and Finance, and founded an AI Capability Center that reaches 1,000+ employees on practical AI tooling, driving a 130% increase in company-wide AI adoption.

• Integrated SharePoint, Outlook, and Claude Cowork into the SVP of Transportation's workflow, lifting delivery tender acceptance rate by 9%.

University of Florida
8 months

Software Development Research Assistant
August 2026 - Present (2 months)
Gainesville, FL

Undergraduate Assistant at UF Grasshopper Lab.

LLM Evaluation Research Assistant
February 2026 - Present (8 months)
Gainesville, FL

As an Undergraduate Research Assistant at the Florida College of Education's Virtual Learning Lab, I contributed to a Multi-Judge LLM Evaluation Pipeline to assess the quality of educational stories, directly impacting 100+ elementary schoolers across Gainesville. This pipeline serves as a core improvement to the Storiza platform.

• Architected a multi-judge LLM evaluation pipeline to assess educational story quality, utilizing a weighted three-model ensemble (Llama 3.1 8B, GPT-OSS 120B, and Nemotron-3-Super 120B) for both rule-by-rule rubric verdicts and holistic scoring.

• Engineered an asynchronous fan-out pipeline using Python's asyncio to execute concurrent API calls, collapsing sequential operations, speeding up grading per story by 500%.

• Implemented statistical quality control by calculating Fleiss' and Cohen's Kappa across judges, creating an automated system to flag rubric rules falling below a κ=0.80 agreement threshold for human revision.

• Benchmarked 11 different language models across small, medium, and large tiers against a human-annotated 19-story ground-truth control set to evaluate cost-to-accuracy tradeoffs for production.

• Enforced deterministic, structured LLM outputs by combining temperature parameter being 0.0 with strict Pydantic JSON schema validation, eliminating malformed response failures.

Dream Team Engineering
Software Engineer
January 2026 - Present (9 months)
Gainesville, FL

At Dream Team Engineering, my team and I are collaborating with a Neurosurgery Resident to develop an application to conduct the Train of Four Test through a mobile phone, providing an alternative to +$3,000 of hardware costs. I designed the app on figma and implemented techniques to detect twitches accurately to calculate the TOF ratio. Currently the app is at a 90% accuracy on Unity simulations. IRB approved trials are upcoming at the UF Shands Hospital.

• Engineered a real-time digital signal processing pipeline utilizing frame differencing, Savitzky-Golay smoothing (15-tap, degree-3 polynomial), and prominence-based peak detection to compute T4/T1 neuromuscular blockade ratios.

• Implemented a Lucas-Kanade pyramid optical flow stabilizer using OpenCV to effectively isolate camera shake from actual hand motion using a background-masked region of interest.

• Designed app UI and wireframes on Figma + Implemented in Kotlin. Implemented Python testing suite to run 400+ videos through 90% accuracy at 5% rate of error.

GatorAI

1 year 2 months

Webmaster

January 2026 - Present (9 months)

Gainesville, FL

Maintaining GatorAI website with 100+ active monthly visitors.

Software Team Lead

August 2025 - December 2025 (5 months)

Gainesville, FL

At GatorAI, I led a team to develop a lightweight AI video generator. It generates videos at $0.05 a minute and so far has been used by 15 unique users. I utilized 5 agents to complete the pipeline and used open source models (Flux-Schnell) to cut costs. I had to come up with a visual bible to provide context for image generation calls as well as utilize seed locking to receive cohesive images. I used Redis and Celery to create message queues to avoid timeouts from calls.

• Designed a 5-agent named state machine (Watchman, Director, Continuity Artist, Auditor, Editor) to orchestrate a complete AI video generation pipeline, tracking granular agent states in PostgreSQL to drive a live frontend dependency graph.

• Engineered a "Global Visual Bible" system utilizing GPT-4o-mini to extract and inject art, color, and lighting styles into Replicate Flux-Schnell prompts alongside deterministic session seeds, ensuring strict cross-slide visual continuity.

• Implemented multi-model concurrent orchestration spanning GPT-4o-mini for structured script generation, Flux-Schnell for image synthesis, and Amazon Polly for text-to-speech, executing parallel workloads.

• Designed an automated Auditor agent featuring per-artifact retry loops to validate image file integrity and audio length, alongside a Watchman pre-flight agent that detects authentication errors early to prevent wasted API spend.

UF Computing Student Union

2 years

Director of User Experience

October 2025 - Present (1 year)

Gainesville, FL

Leading and reviewing designs and contributing to mockups.

User Experience Designer
October 2024 - October 2025 (1 year 1 month)
Gainesville, FL

At UF Computing Student Union, my team and I worked on making designs, components, and design system on figma. The club platforms currently serve 300+ students on campus and collaborate with 25+ computing clubs on campus. My primary work was in developing designs for mobile and computer sites and implementing the design system on React

• Contributed to the UX/UI design of a centralized club executive platform, UF club directory, and official CSU site in Figma, successfully driving adoption across 300+ student leaders and 20+ organizations university-wide.

• Implementing design system on React comprising 200+ interactive components, standardizing global typography, color palettes, and UI assets to ensure visual consistency and accelerate developer handoff.

• Conducted competitive analysis and user research to inform the platform's information architecture, mapping out intuitive user journeys for complex club management workflows.

• Executed 15+ rapid design iterations driven by A/B testing and direct user feedback, continuously optimizing interface layouts to improve usability.

• Designed high-fidelity, interactive prototypes to validate accessibility standards and user flows, effectively bridging the gap between user-centric design concepts and technical implementation.

Swamp Records
Frontend Developer
January 2026 - April 2026 (4 months)
Gainesville, FL

At Swamp Records, my team and I implemented a digital portfolio/platform for Gainesville artist Alex Willow. He has ~50+ listeners on spotify. I designed the pages on figma and then later implemented some of them on React. Now the site is deployed live for all Fans of Alex Willow.

• Designed on Figma and, developed, deployed an artist portfolio and music platform for Alex Willow under Swamp Records, leveraging Next.js and React.

• Designed a clean React Server Components architecture with a strict Server/Client boundary to separate static layouts and metadata from interactive client-side components.

• Developed a fully responsive, multi-breakpoint UI design utilizing Tailwind CSS v4's modern CSS-first workflow and fluid fluid typography layout techniques

Gator User Design
User Experience Designer
August 2025 - January 2026 (6 months)

At Gator User Design I worked with a team to redesign the UFCFSG lab website. Collaborating with the professor we took feedback and made around 3 drafts after conducting competitive analysis, forming user personas and conducting A/B testing.

• Designed the end-to-end UX/UI redesign of the UF Computing For Social Good Lab website, delivering a finalized high-fidelity prototype that was officially accepted and is now in active development.

• Conducted extensive competitive analysis across top-tier university research lab websites to inform the information architecture and establish modern usability standards.

• Collaborated directly with the lab's lead professor and key stakeholders to gather continuous feedback, translating complex academic and research requirements into intuitive user journeys.

• Validated layout and navigation decisions through structured A/B testing, optimizing the user experience to effectively highlight the lab's core research initiatives, publications, and impact.

English2Success
Software Engineering Intern
February 2025 - July 2025 (6 months)
Lake Mary, FL

At English2Success I completely replaced a manual process of gathering progress from tutors into a automated dashboard. It saved the organization 8 hours a week and helped expand its members from 200 to 450 global users. I used Flask and GCP to automate form creation/distribution and then intaking the response/summarizing them and displaying it to the correct people.

• Architected and deployed a full-stack progress reporting dashboard utilizing a Flask REST API, PostgreSQL, and a custom HTML/CSS/JavaScript frontend, successfully scaling the platform from 200 to 450 active users.

• Engineered an automated data ingestion pipeline leveraging GCP and Google Forms to schedule weekly form distributions and securely capture user response data for real-time tracking.

• Integrated the OpenAI API (GPT) to automatically summarize and validate incoming form responses, implementing an AI-driven anomaly detection layer to ensure data integrity without manual review.

• Optimized backend database performance by designing indexed virtual tables (materialized views) to accelerate complex SQL joins, reducing query latency and improving dashboard load speeds by 15%.

Education

University of Florida

Bachelor of Science · (August 2024 - May 2028)